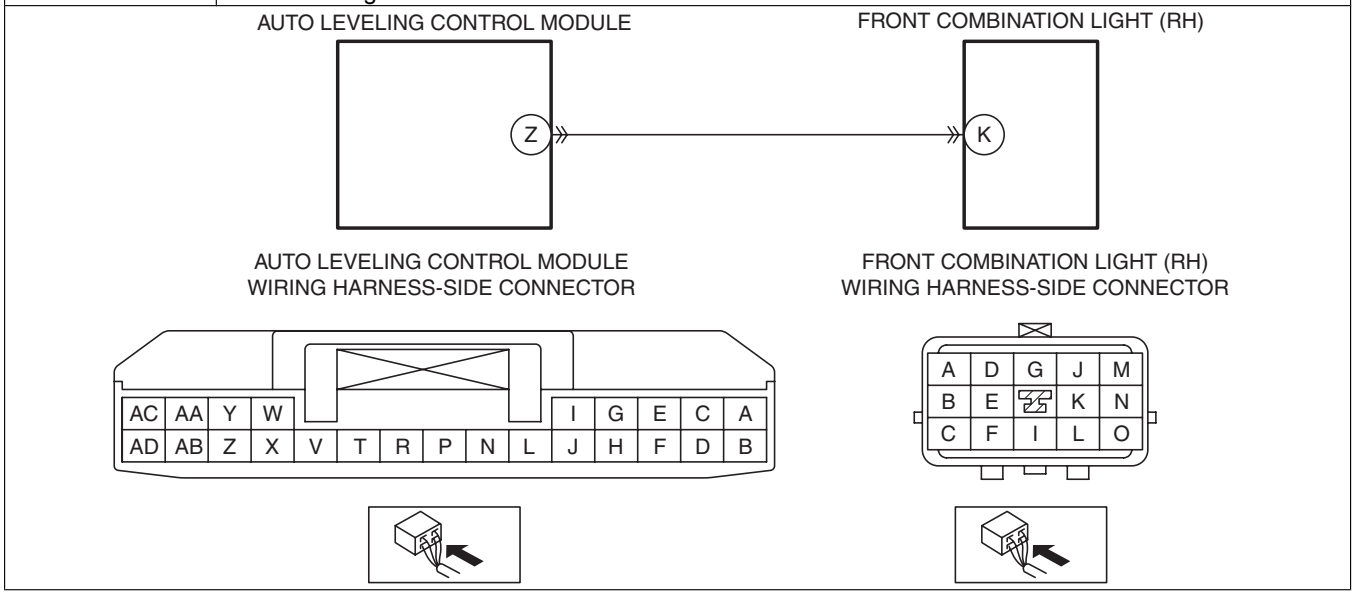
DTC B1D01:13 [HEADLIGHT AUTO LEVELING SYSTEM]

id0902i7060100

Description	Headlight (RH) circuit malfunction
Detection condition	• The auto leveling control module detected an open circuit in the headlight (RH) circuit during the headlight LO on control with the ignition switched ON (engine off or on).
Fail-safe	• Control of the headlight auto leveling system is continued.
Possible cause	• DTCs are stored in the electrical supply unit (ESU). — Headlight LO (RH) circuit malfunction • Front combination light (RH) connector or terminal malfunction • Auto leveling control module connector or terminal malfunction • Short to ground in wiring harness between auto leveling control module terminal Z and front combination light (RH) terminal K • Open circuit in wiring harness between auto leveling control module terminal Z and front combination light (RH) terminal K • Front combination light (RH) malfunction • Auto leveling control module malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY ELECTRICAL SUPPLY UNIT (ESU) DTCs** • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].)
		No	Go to the next step.
2	**INSPECT FRONT COMBINATION LIGHT (RH) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the front combination light (RH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.

Step	Inspection		Action
3	**INSPECT AUTO LEVELING CONTROL MODULE CONNECTOR CONDITION** • Disconnect the auto leveling control module connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
4	**INSPECT HEADLIGHT (RH) MALFUNCTION DETECTION CIRCUIT FOR SHORT TO GROUND** • Verify that the front combination light (RH) and auto leveling control module connectors are disconnected. • Inspect for continuity between front combination light (RH) terminal K (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between auto leveling control module terminal Z and front combination light (RH) terminal K. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
5	**INSPECT HEADLIGHT (RH) MALFUNCTION DETECTION CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition off. • Verify that the front combination light (RH) and auto leveling control module connectors are disconnected. • Inspect the wiring harness for continuity between auto leveling control module terminal Z (wiring harness-side) and front combination light (RH) terminal K (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between auto leveling control module terminal Z and front combination light (RH) terminal K. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
6	**VERIFY IF MALFUNCTIONING LOCATION IS FRONT COMBINATION LIGHT (RH) DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the auto leveling control module using the M-MDS. (See CLEARING DTC [HEADLIGHT AUTO LEVELING SYSTEM].) • Operate the light switch to the HEAD position and the dimmer switch to the LO position. • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the auto leveling control module DTCs using the M-MDS. (See DTC INSPECTION [HEADLIGHT AUTO LEVELING SYSTEM].) • Is the same DTC displayed?	Yes	Replace the front combination light (RH), then go to the next step. (See FRONT COMBINATION LIGHT REMOVAL/ INSTALLATION.)
		No	Go to Step 8.

Step	Inspection		Action
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the auto leveling control module using the M-MDS. (See CLEARING DTC [HEADLIGHT AUTO LEVELING SYSTEM].) • Operate the light switch to the HEAD position and the dimmer switch to the LO position. • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the auto leveling control module DTCs using the M-MDS. (See DTC INSPECTION [HEADLIGHT AUTO LEVELING SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the auto leveling control module. (See AUTO LEVELING CONTROL MODULE REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [HEADLIGHT AUTO LEVELING SYSTEM].)
		No	DTC troubleshooting completed.